

Introduction to Soft Computing

Unit 1: Computing Systems, Soft vs. Hard Computing, Characteristics & Applications

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Soft Computing (CSA301) — Unit 1

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Course Page:

<https://www.gcjana.in/courses/shardauniversity/2601/soft-computing/>

Outline

- 1 Concept of Computing Systems
- 2 What is Soft Computing?
- 3 Soft Computing versus Hard Computing
- 4 Characteristics of Soft Computing
- 5 Applications of Soft Computing
- 6 Summary and Practice

Learning Outcomes of Unit 1

After this unit, a student will be able to...

- ➊ **Explain** what a computing system is and what its classical assumptions are.
- ➋ **Define** Soft Computing and state Zadeh's guiding principle.
- ➌ **Distinguish** Soft Computing from Hard Computing with examples.
- ➍ **Identify** the constituent paradigms — Fuzzy Logic, Neural Networks, Evolutionary Computing, Probabilistic Reasoning.
- ➎ **List and justify** the characteristics of Soft Computing.
- ➏ **Map** real-life problems to the appropriate Soft Computing technique.

Mapping to the Syllabus

A. Concept of computing systems. What is Soft Computing? Characteristics and applications. **B.** “Soft” vs. “Hard” Computing. **C.**

What is Computing?

Definition

Computing is the process of **mapping** some input information space into an output space, using a well-defined procedure (an **algorithm**).

- Input $\rightarrow$ Computing System $\rightarrow$ Output
- The mapping $y = f(x)$ must be **precisely specified**.
- Also known as **Hard Computing** or **Conventional Computing**.

Everyday examples

Sorting a list, computing a bank balance, solving $ax^2 + bx + c = 0$, rendering a spreadsheet.

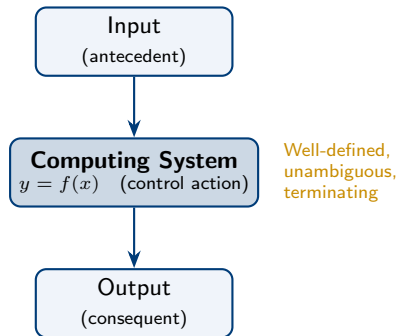


Figure: The classical computing paradigm.

Important Characteristics of (Classical) Computing

Three non-negotiable requirements

- ① **Precise control action:** the control action must be *unambiguous* and *accurate*, expressible as an **analytical formulation** (a mathematical model / algorithm).
- ② **Exactness:** for a given input the system must always produce the *same, exact* output — the answer is either right or wrong.
- ③ **Certainty:** inputs must be complete, noise-free and unambiguous.

Control action

A guiding rule the system *must* follow, e.g.
IF marks ≥ 90 THEN grade = EX

Consequence

No formulation $\Rightarrow$ no program.
Vague input $\Rightarrow$ undefined behaviour.

A Classical Computing System at Work

Problem: Grade a student

Marks	Grade
≥ 90	EX
$80 \leq M < 90$	A
$70 \leq M < 80$	B
< 70	F

- Model is **exact**; boundaries are **sharp**.
- $79.9 \Rightarrow B$, $80.0 \Rightarrow A$.
- Nothing is left to interpretation.

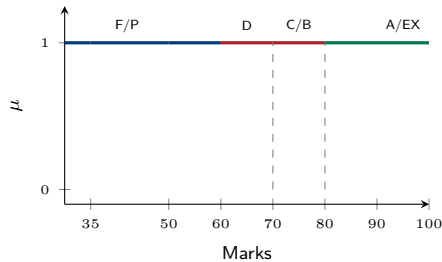


Figure: Crisp (binary) membership — the hallmark of hard computing.

Why Can't Computers Do *Everything*?

Limitations of classical computing

- Algorithms assume **exact and complete** information.
- Cannot handle **partial** or **approximate** knowledge.
- Real-world information is inherently **vague**.
- Many problems have **no analytical model** at all.
- Some models exist but are **computationally intractable** (NP-hard).

Typical problem areas

- Hospital information management, medical diagnosis
- Public administration / policy systems
- Flood, fire and disaster control systems
- Handwriting, speech and face recognition
- Driving a car in Delhi traffic
- Weather forecasting

A human does these effortlessly — without solving a single differential equation.

Class Engagement: Can You Write the Algorithm?

Task for you (2 minutes)

Write a **precise, exact** rule for:

- 1 "Is this person *tall*?"
- 2 "Is the tea *hot*?"
- 3 "Park this car in that gap."
- 4 "Is this handwritten symbol a '2'?"

What usually happens

- *Tall*: is 5 ft 11.9 in tall but 6 ft 0.0 in is? **Boundary problem.**
- *Hot*: depends on person, cup, season. **Context problem.**
- *Parking*: no equations used by any driver, yet everybody succeeds. **Model-free skill.**
- *Digit '2'*: infinitely many shapes. **Pattern problem.**

Humans reason with **words**, not numbers — and still achieve **robust** results at **low cost**.

What is Soft Computing?

Definition (L.A. Zadeh)

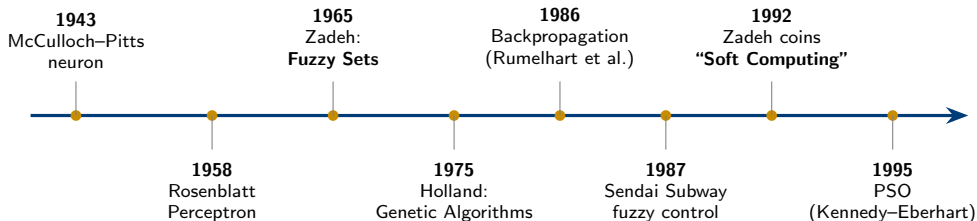
*"Soft computing differs from conventional (hard) computing in that, unlike hard computing, it is **tolerant of imprecision, uncertainty, partial truth and approximation**. The role model for soft computing is the **human mind**."*

The Guiding Principle

*Exploit the tolerance for imprecision, uncertainty, partial truth and approximation to achieve **tractability, robustness, low solution cost** and better rapport with reality.*

- Soft computing is **not** a single technique — it is a **consortium** / **partnership** of methodologies.
- The members are **complementary**, not competitive.

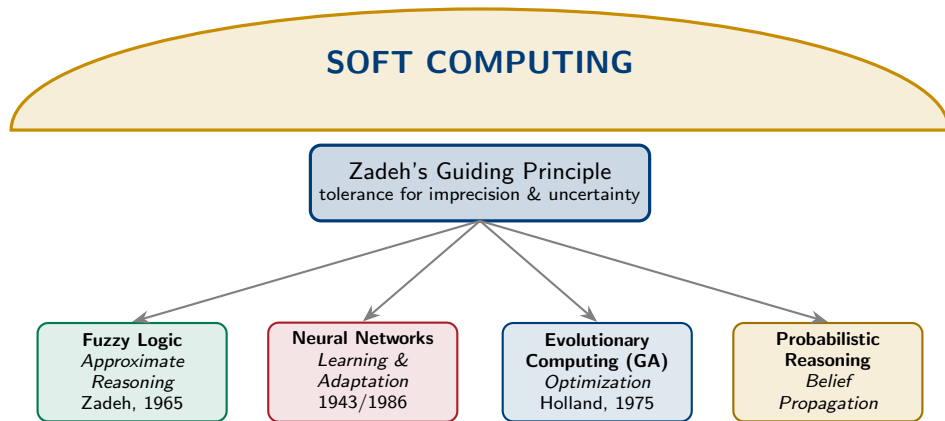
A Brief History — Milestones



Note

Fuzzy Logic was introduced by **Lotfi A. Zadeh** (UC Berkeley, 1965) — the *father of fuzzy logic*, fondly nicknamed **LAZ**. He later coined the term *Soft Computing* in the early 1990s and founded **BISC** (Berkeley Initiative in Soft Computing, 1991).

The Soft Computing Umbrella



+ Swarm Intelligence (ACO/PSO), Chaos Theory, Rough Sets, Machine Learning

Figure: Soft computing is an *umbrella* of complementary, biologically inspired paradigms.

Constituent 1 — Fuzzy Logic: Handling *Imprecision*

Idea

Replace binary membership $\{0, 1\}$ with a **degree of membership** $\mu_A(x) \in [0, 1]$.

$$A = \{(x, \mu_A(x)) \mid x \in X\}$$

Crisp vs. Fuzzy

Crisp: Is the liquid colourless?	Water = Yes, Milk = No
Fuzzy: Is the person honest?	Ankit = 0.99, Kabita = 0.35

- Knowledge is captured as **IF-THEN** rules in words.
- *IF temperature is cool THEN fan is slow.*

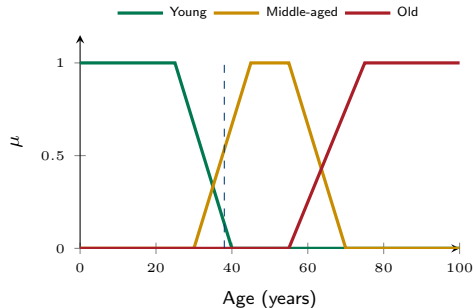


Figure: A 38-year-old is *young* to degree 0.13 and *middle-aged* to degree 0.53 — overlap is allowed!

Constituent 2 — Neural Networks: Learning from *Data*

Idea

Mimic the brain: many simple **neurons** connected by **weighted** links that are **learned** from examples.

$$y = \varphi \left(\sum_{i=1}^n w_i x_i + b \right)$$

- **Model-free**: no equation is supplied by the designer.
- Learns by **adjusting weights** (backpropagation).
- Excellent for **pattern recognition**, classification, approximation.
- Human brain: $\sim 10^{11}$ neurons, $\sim 10^{15}$ synapses.

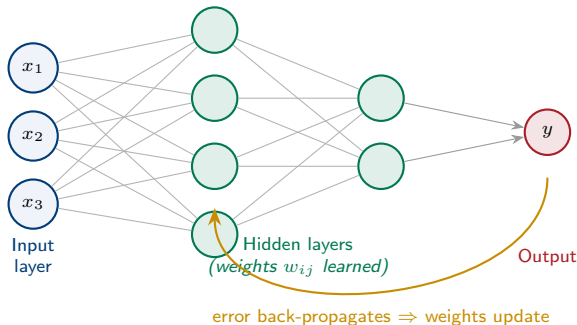


Figure: A feed-forward artificial neural network.

Constituent 3 — Genetic Algorithms: *Optimization* by Evolution

Idea (Holland, 1975)

Imitate **Darwinian evolution**: maintain a *population* of candidate solutions, and let **survival of the fittest** drive the search.

- **Encoding**: solution $\rightarrow$ chromosome, e.g. 1 0 1 1
0 0 1
- **Selection**: fitter parents reproduce more
- **Crossover**: 1011|001 $\times$ 0100|110 $\rightarrow$ 1011110
- **Mutation**: random bit flip $\Rightarrow$ diversity

Why it matters

Finds *good enough* solutions to **NP-hard** problems (TSP, scheduling, routing) where exhaustive search is impossible.

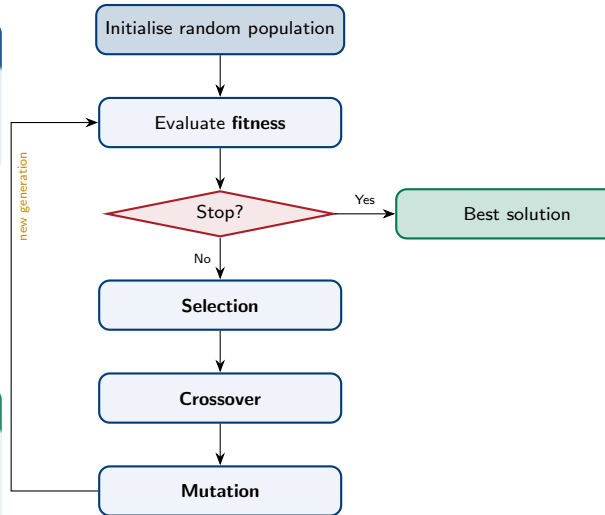


Figure: The basic GA framework (detailed in Unit 4).

Constituent 4 — Probabilistic Reasoning & Swarm Intelligence

Probabilistic Reasoning

Handles **uncertainty** / **randomness** (not vagueness).

$$P(H \mid E) = \frac{P(E \mid H) P(H)}{P(E)}$$

- Bayesian networks, belief propagation
- Dempster–Shafer evidence theory
- Hidden Markov Models, chaos theory

Remember

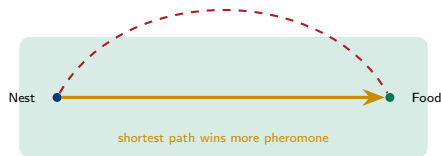
Fuzzy → “the patient has a **mild** fever” — the *category* is vague.
Probability → “India wins with 60% chance” — the outcome is crisp, the *chance* is random.

Swarm Intelligence

Collective intelligence emerging from **simple agents** following **simple local rules**.

- **ACO** (Dorigo, 1992) — ants + pheromone → shortest path
- **PSO** (Kennedy & Eberhart, 1995) — bird flocking

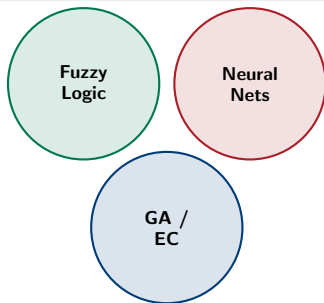
$$v_i \leftarrow wv_i + c_1r_1(p_i - x_i) + c_2r_2(g - x_i)$$



Hybrid Soft Computing Systems

Zadeh's key insight

The constituents of soft computing are **complementary rather than competitive**. Their real power appears in **hybrid** (synergistic) combinations.



Hybrid	What it buys you
Neuro-Fuzzy (ANFIS)	rules + learning (Unit 3)
Fuzzy-GA	auto-tuned membership functions
Neuro-GA	evolved network topology/weights
Fuzzy-Neuro-GA	full self-tuning controller

Example

A **neuro-fuzzy** washing machine: the fuzzy rule base gives *interpretability*; the neural layer *learns* the membership functions from sensor data — the best of both worlds.

Quick Check — Round 1

[Q1] Who coined the term *Soft Computing*, and in which decade?

[Q2] State Zadeh's guiding principle in one sentence.

[Q3] Which paradigm would you choose for each?

- 1 Encoding a doctor's experience as rules
- 2 Recognising handwritten pin-codes from 60,000 samples
- 3 Finding a near-optimal delivery route over 100 cities

[Q4] True/False: Soft computing is a *single* algorithm that replaces hard computing.

Think for 60 seconds — answers on the next slide.

Answers — Round 1

[A1]

Lotfi A. Zadeh, in the **early 1990s** (he had introduced fuzzy sets much earlier, in 1965).

[A2]

Exploit the tolerance for imprecision, uncertainty, partial truth and approximation to achieve tractability, robustness, low solution cost and better rapport with reality.

[A3]

(i) **Fuzzy Logic** — knowledge is linguistic. (ii) **Neural Networks** — learning from labelled data. (iii) **Genetic Algorithm** — combinatorial optimisation (TSP is NP-hard).

[A4]

False. It is a **consortium/partnership** of complementary methodologies, and it **complements** hard computing rather than replacing it.

Hard Computing — The Classical Way

Premises of Hard Computing

- 1 A **precise analytical model** exists.
- 2 The computation is **deterministic**: same input $\Rightarrow$ same output.
- 3 The answer must be **exact and provably correct**.
- 4 Programs are **sequential** in logic.
- 5 Data must be **complete** and **noise-free**.

Strengths

Precision, verifiability, reproducibility, mathematical guarantees.

Classic hard-computing tools

- Binary logic, crisp sets
- Numerical analysis, PID control
- Deterministic search / sorting
- Symbolic AI, expert rules (crisp)

Weaknesses

- Brittle — fails badly on noisy/incomplete data
- Needs a model that may not exist
- Expensive (time, memory) for NP-hard problems

Soft Computing — The Human-Inspired Way

Premises of Soft Computing

- ① The real world is **pervasively imprecise and uncertain**.
- ② Precision **costs** — and is often unnecessary.
- ③ **Tolerance** for imprecision can be *exploited* to get tractability, robustness and low cost.
- ④ The role model is the **human mind**.

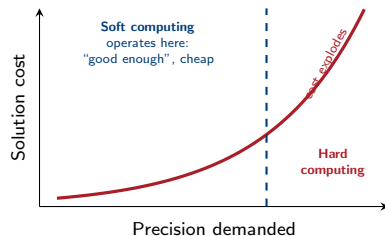


Figure: The *precision–cost* trade-off. Soft computing deliberately trades a little accuracy for a lot of savings.

“Precision and certainty carry a cost.” — L.A. Zadeh

Hard Computing vs. Soft Computing — Comparison

Aspect	Hard Computing	Soft Computing
Model	Requires a precise analytical model	Model-free ; tolerant of no/partial model
Data	Exact, complete, noise-free	Imprecise, noisy, incomplete, ambiguous
Logic	Binary / two-valued	Multi-valued, fuzzy, probabilistic
Computation	Deterministic, sequential	Stochastic, parallel & distributed
Result	Exact, guaranteed	Approximate , “good enough”
Behaviour on noise	Brittle — may fail completely	Robust — graceful degradation
Programming	Explicitly programmed	Learns / evolves from data & experience
Cost	High for complex problems	Low solution cost
Role model	The Turing machine	The human mind
Example	PID controller, sorting	Fuzzy AC, ANN face recogniser, GA scheduler

They are complements, not rivals: many real systems use hard computing inside a soft-computing wrapper.

Illustration — Parking a Car (Zadeh's Favourite Example)

Hard computing approach

- Model the vehicle kinematics exactly.
- Solve for trajectory with ± 1 cm tolerance.
- Needs sensors, differential equations, heavy computation.
- **High cost**, and still fails if the gap is odd-shaped.

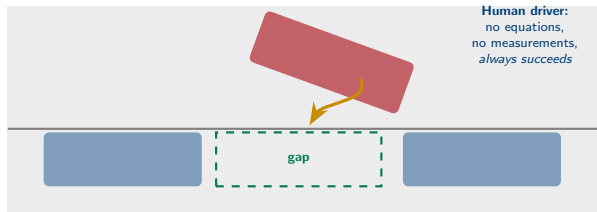


Figure: Parallel parking — an *ill-defined* problem that humans solve with *imprecise, linguistic* reasoning.

Soft computing approach

*IF gap is **roughly** car-length AND angle is **about** 40° THEN turn **sharply** right.*

Moral: the tolerance for imprecision is not a bug — it is the *resource* that makes the problem tractable.

Illustration — “Reaching Home from College”

Hard computing answer

$$\text{“Travel time} = \frac{d}{v} = \frac{12.4 \text{ km}}{31.2 \text{ km/h}} = 23.85 \text{ minutes.”}$$

- Assumes constant speed, no signals, no jam.
- Precise ... and **almost always wrong**.

Soft computing answer

“It will take **about half an hour, maybe more** if traffic is **heavy**.”

- Imprecise ... and **practically useful**.
- Robust to the unmodelled real world.

More everyday contrasts

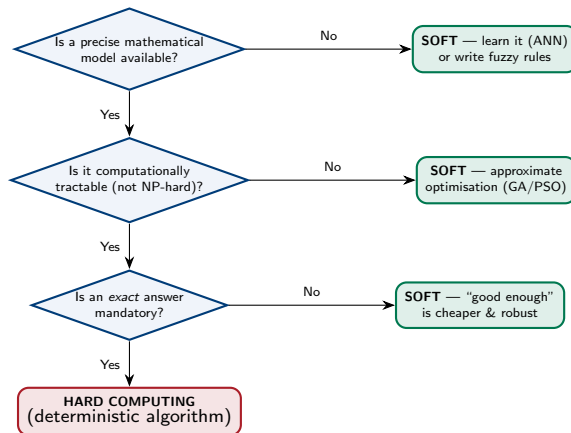
Task	Which wins?
Bank interest	Hard (must be exact)
Rocket trajectory	Hard (model known)
Face recognition	Soft (no model)
Stock prediction	Soft (noisy, non-linear)
Medical diagnosis	Soft (vague symptoms)
Traffic signals	Soft (dynamic, uncertain)
Payroll	Hard (exactness is legal)

Rule of thumb

Ask: *Do I have a model?* and *Is exactness essential?*

Both yes $\Rightarrow$ hard. **Any no** $\Rightarrow$ consider soft.

Decision Guide — Which One Do I Use?



Best practice

In industry the two are **combined**: e.g. a hard-computed safety interlock guarding a fuzzy controller, or a GA that *tunes* the gains of a classical PID loop.

Quick Check — Round 2

[Q1] Fill in the blanks

Hard computing needs a _____ model, produces _____ results and is _____ to noise, whereas soft computing is _____-free, produces _____ results and degrades _____.

[Q2]

Your bank must compute monthly EMI. Would you use soft computing? Justify.

[Q3]

A washing machine must decide the wash time from the *dirtiness* and *type* of the load. Hard or soft? Why?

[Q4]

Explain in one line why “precision costs”.

Answers — Round 2

[A1]

precise/analytical · exact · brittle/sensitive model · approximate · gracefully.

[A2]

No — use hard computing. The model $EMI = \frac{P r (1+r)^n}{(1+r)^n - 1}$ is exact and known, and legal/financial correctness demands an exact answer. “Approximately Rs. 12,000” is unacceptable.

[A3]

Soft (fuzzy). “Dirtiness” is a *vague, linguistic* variable with no analytical model; sensor data is noisy; an *approximately right* wash time is perfectly acceptable. Rules like *IF dirt is high AND load is cotton THEN wash time is long* capture the expert’s knowledge directly.

[A4]

Every extra digit of accuracy demands better sensors, finer models and more computation — cost grows steeply while the *usefulness* of the answer often does not.

Characteristics of Soft Computing — Overview

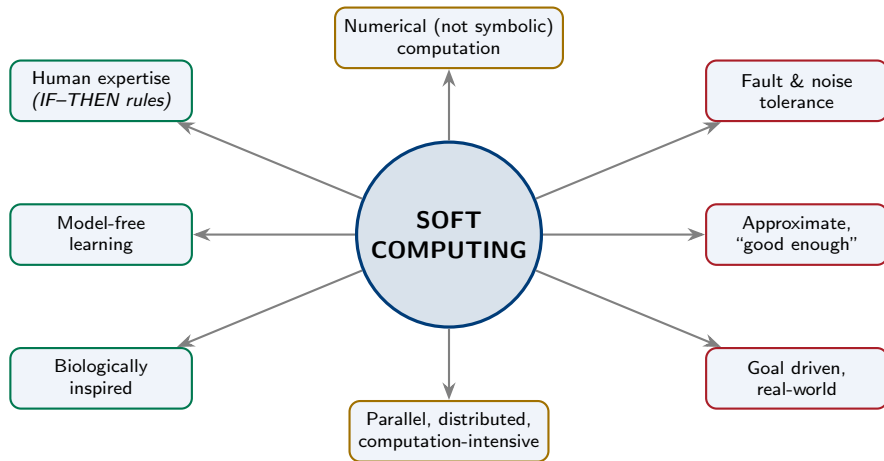


Figure: The eight defining characteristics, unpacked over the next slides.

Characteristic 1–3

1. Human Expertise

Soft computing systems use **human knowledge** in the form of **fuzzy IF–THEN rules** and human-like reasoning, instead of a designer-supplied equation. *Example:* a plant operator's rule "*if the furnace is a bit too hot, cut the fuel slightly*" becomes a fuzzy rule directly.

2. Model-Free Learning

ANNs and adaptive systems **construct the input–output mapping from data alone**. No physical/mathematical model of the plant is required. *Example:* predicting electricity demand from 5 years of load history.

3. Biologically Inspired Computing Models

The design metaphors come from nature:

- Brain → **Neural Networks**
- Evolution/genetics → **Genetic Algorithms**
- Ant colonies / bird flocks → **ACO / PSO**
- Human linguistic reasoning → **Fuzzy Logic**

Characteristic 4–6

4. New Optimisation Techniques

Derivative-free, **stochastic global search** (GA, SA, PSO, ACO) that works where gradient methods fail: discontinuous, multi-modal or combinatorial landscapes.

5. Numerical Computation

Unlike symbolic AI, soft computing is fundamentally **numerical** — membership grades, weights, fitness values.

6. New Application Domains

Because no model is needed, entirely new domains become programmable: appliances, bioinformatics, finance, image processing.

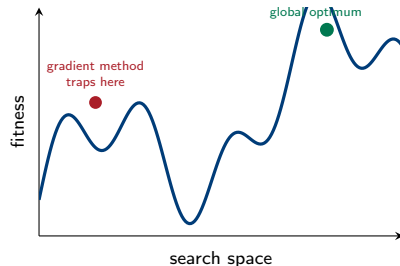


Figure: A multi-modal landscape — classical (hard) optimisation gets stuck in a *local* optimum; population-based soft methods explore globally.

Characteristic 7–8: Fault Tolerance & Goal-Driven Behaviour

7. Fault & Noise Tolerance

- Knowledge is **distributed** across many units/rules — no single point of failure.
- Damage to a few neurons/rules causes only a **graceful degradation** in performance, not a crash.
- Contrast: one wrong line in a C program $\Rightarrow$ total failure.

8. Goal-Driven & Real-World Oriented

The path to the solution is **irrelevant**; only reaching the goal matters. Soft computing is defined by the problems it solves, not by the theorems it proves.

Cost of these benefits: soft computing is *computation-intensive* and gives *no proof of optimality*.

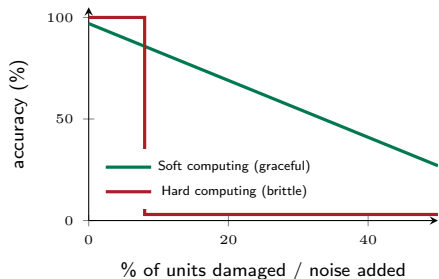


Figure: *Graceful degradation* — the signature behaviour of soft computing systems.

Summary Card: Characteristics of Soft Computing

Characteristic	Meaning / Consequence
Human expertise	Knowledge encoded as linguistic IF–THEN rules
Model-free learning	Mapping is learned from data, not derived
Biological inspiration	Brain, evolution, swarms as design metaphors
Approximate solution	“Good enough” beats “exact but impossible”
Tolerance	Accepts imprecision, uncertainty, partial truth
Robustness	Graceful degradation under noise and faults
Parallel/distributed	Naturally suited to parallel hardware/GPU
Adaptivity	Self-tunes as the environment changes
Numerical	Operates on grades, weights, fitness values
Low solution cost	Cheap sensors, cheap computation, fast deployment
Goal driven	Only the outcome matters, not the derivation
Limitations: no optimality proof; needs data or an expert; results may vary run to run; tuning is empirical.	

Quick Check — Round 3

[Q1]

“Soft computing systems show *graceful degradation*.” Explain what this means and why hard computing does not.

[Q2]

Which characteristic of soft computing makes it suitable when *no mathematical model* of the plant exists?

[Q3] MCQ

Which is *not* a characteristic of soft computing?

(a) Model-free learning (b) Guaranteed optimal solution (c) Fault tolerance (d) Biological inspiration

[Q4]

Name two costs/limitations you accept when you choose soft computing.

[A1]

Knowledge is stored **redundantly and in a distributed way** across many weights/rules, so losing a few of them lowers accuracy *a little* instead of stopping the system. In hard computing the logic is a *single exact chain*: break one link and the whole computation is wrong.

[A2]

Model-free learning (supported by *human expertise* when rules are available instead of data).

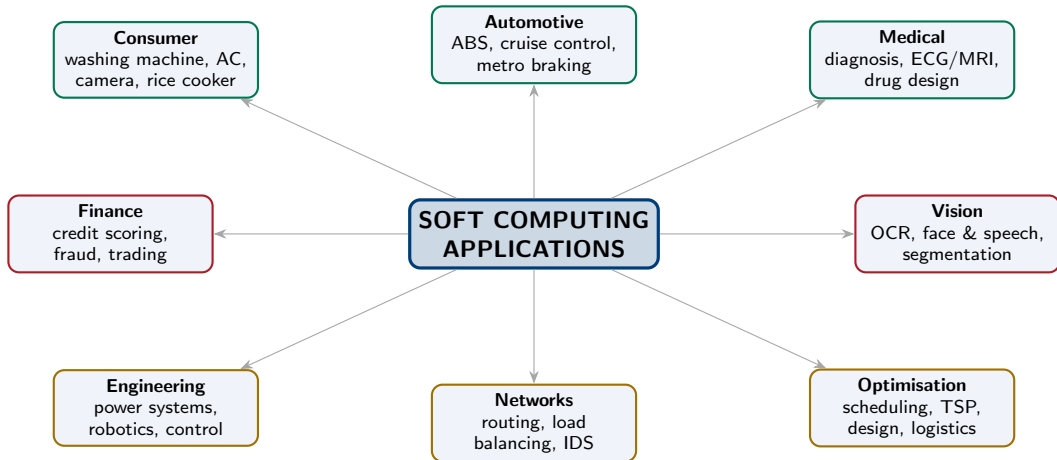
[A3]

(b) Guaranteed optimal solution. Soft computing returns an *approximate, near-optimal* answer with no proof of optimality.

[A4]

(i) Computationally intensive / slow training; (ii) no guarantee of optimality or repeatability; (iii) needs representative data or a good expert; (iv) empirical tuning (any two).

The Application Landscape



Application Domain 1 — Consumer Appliances

Fuzzy logic in your home

- **Washing machine** (Matsushita, 1990): senses load size, fabric and dirtiness → sets water level, detergent, wash & rinse time.
- **Air conditioner**: smooth compressor control → ~20–30% energy saving vs. on-off control.
- **Rice cooker**: adjusts cooking by rice quantity and moisture.
- **Camera / camcorder**: fuzzy auto-focus, stabilisation.
- **Microwave, vacuum cleaner, fridge**: adaptive power.

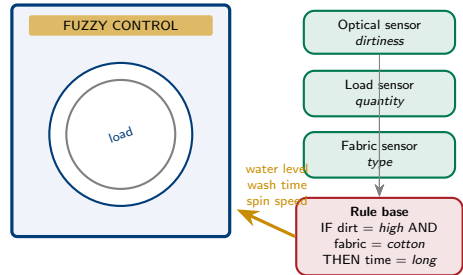


Figure: Sensors → fuzzy rule base → actuators.

Why fuzzy and not PID?

Cheap sensors, no plant model, human-defined comfort — the customer only needs “clothes come out clean”, not $\pm 0.1\%$ accuracy.

Application Domain 2 — Transportation & Automotive

Landmark: Sendai Subway, Japan (1987)

Hitachi's **fuzzy automatic train operation** controlled acceleration and braking of the Sendai metro:

- **Smoother** rides than human drivers (fewer jerks)
- ~ 10% **lower energy** consumption
- More **accurate** stopping at platforms

The first large-scale public proof that soft computing works in safety-critical engineering.

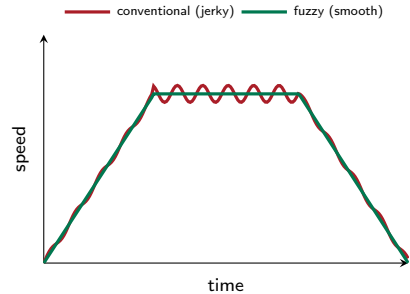


Figure: Fuzzy train control produces a smooth speed profile —
comfort ↑, energy ↓, wear ↓.

Other automotive uses

- Fuzzy ABS & traction control, automatic transmission
- Fuzzy/neural engine & emission control
- ANN-based driver-drowsiness and lane detection
- GA/ACO for traffic-signal timing and fleet routing

Application Domain 3 — Medicine, Bioinformatics & Vision

Medical & biological

- **Fuzzy diagnosis**: symptoms are inherently vague — “*mild* fever”, “*severe* pain”. Rule bases mirror the clinician’s reasoning and remain **explainable**.
- **ANN**: tumour detection in MRI/CT, ECG arrhythmia classification, diabetic-retinopathy screening.
- **GA**: protein folding, drug design, gene selection from microarrays, radiotherapy dose planning.
- **Neuro-fuzzy**: anaesthesia depth control, ICU alarm management.

Image & signal processing

- Handwriting / OCR (pin-code sorting by India Post-style systems)
- Face, iris and fingerprint recognition
- Fuzzy edge detection, contrast enhancement, denoising
- Speech recognition, speaker identification
- Image compression and segmentation

Student takeaway

Every one of these has **no clean equation**, **noisy input** and a **human notion of “correct”**. That is exactly the soft-computing signature.

Application Domain 4 — Finance, Industry, Networks & Optimisation

Business & finance

- Credit scoring & loan approval (fuzzy + ANN)
- Fraud/anomaly detection in transactions
- Stock-market & exchange-rate prediction
- Portfolio optimisation (GA, NSGA-II — Unit 5)
- Customer segmentation, demand forecasting

Networks & computing

- ACO-based routing, load balancing
- Intrusion-detection systems (ANN/fuzzy)
- Cloud resource allocation & VM placement (GA/PSO)
- Wireless sensor-network clustering

Power & process industry

- Load forecasting, economic load dispatch (GA/PSO)
- Fuzzy control of cement kilns, steel furnaces
- Fault diagnosis in transformers and motors
- Renewable-energy MPPT control

Classical optimisation problems

- Travelling Salesman, vehicle routing
- Job-shop & timetable scheduling
- Knapsack, bin packing, VLSI floor-planning
- Structural & antenna design

Case Study A — Fuzzy Air Conditioner (Walk-Through)

Rule base (2 of many)

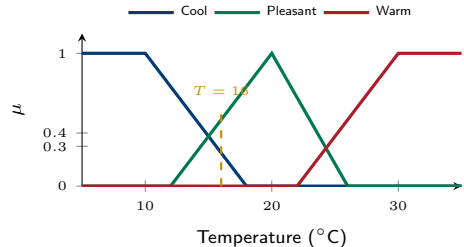
R1: IF temperature is **Cool** THEN fan is **Slow**

R2: IF temperature is **Pleasant** THEN fan is **Medium**

Input: $T = 16^{\circ}\text{C}$

- 1 **Fuzzify**: $\mu_{\text{Cool}}(16) = 0.3$, $\mu_{\text{Pleasant}}(16) = 0.4$
- 2 **Infer**: R1 fires at 0.3, R2 fires at 0.4 (min/product)
- 3 **Compose**: aggregate the clipped outputs (max)
- 4 **Defuzzify**: centre of gravity $\eta = \frac{\sum y \mu(y)}{\sum \mu(y)} \Rightarrow$ crisp fan speed

Full treatment in Units 2–3.



crisp

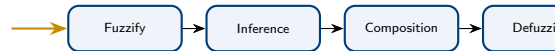


Figure: The fuzzy inference pipeline.

Case Study B — GA for the Travelling Salesman Problem

Why hard computing fails

For n cities there are $(n-1)!/2$ tours.

$n = 20 \Rightarrow 6 \times 10^{16}$ tours.

At 10^9 tours/second $\Rightarrow \approx 1.9$ years.

$n = 30 \Rightarrow$ longer than the age of the universe.

Soft computing (GA) solution

- **Chromosome**: a permutation, e.g. D-K-M-C-B-...
- **Fitness**: $1/\text{tour length}$
- **Operators**: order crossover, swap mutation
- **Result**: a tour within a few % of optimal in **seconds**

Trade: give up the *proof* of optimality, gain a *usable* answer.

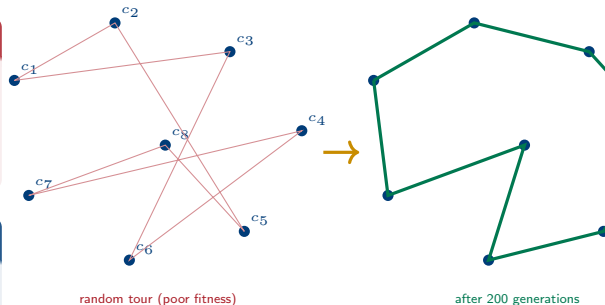


Figure: Evolution improves the tour generation by generation.

Case Study C — ANN for Handwritten Digit Recognition

The problem

Write the *exact* rule that separates a handwritten '2' from a '7'. Nobody can — yet every human reads them instantly.

The soft computing solution

- Input: $28 \times 28 = 784$ pixel intensities
- Network learns weights from 60,000 labelled samples
- Output: 10 confidence values, e.g. $P(2) = 0.93$, $P(7) = 0.05$
- Accuracy $> 99\%$ — **robust** to smudges, slant, thickness

Characteristics on display

model-free learning · biological inspiration · approximation · noise tolerance · numerical computation

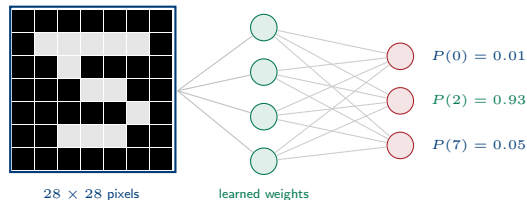


Figure: Pixels in, *degrees of confidence* out — an approximate answer that is right 99% of the time.

Tools You Will Use in This Course (Self-Learning: 30 h)

Python toolboxes

Library	Paradigm
scikit-fuzzy	Fuzzy sets, FIS, controllers
DEAP	Genetic Algorithms / EC
pyswarms	Particle Swarm Optimisation
pymoo	NSGA-II, multi-objective
TensorFlow/Keras	Neural networks

Self-learning components

- Coding (15 h) · Video-based learning (15 h)
- Numerical assignments (10 h) · Case study (8 h)

Road map of CSA301

- **Unit 1** — Introduction to Soft Computing (you are here)
- **Unit 2** — Fuzzy Logic: sets, operations, relations, defuzzification
- **Unit 3** — Fuzzy Inference Systems: Mamdani, Sugeno, Tsukamoto, ANFIS
- **Unit 4** — Genetic Algorithms, ACO, PSO
- **Unit 5** — Multi-objective optimisation, MOEA, Pareto approaches

Assessment reminder

CA 25 (Assessment-1 on Units 1–2) + MSE 15 + ESE 60 = 100.

Limitations & Honest Caveats

What soft computing cannot promise

- **No optimality proof** — only “good enough”.
- **Non-repeatable**: stochastic methods may give different answers per run.
- **Data hungry** (ANN) or **expert hungry** (fuzzy).
- **Empirical tuning**: MF shapes, learning rate, population size.
- **Computationally intensive** training/search.
- **Opacity**: ANNs are black boxes (fuzzy rules are not).

So, when should you *not* use it?

- Safety-critical exactness (avionics arithmetic, cryptography)
- Legal/financial computation (interest, tax, payroll)
- Problems with a known, cheap, exact model

The mature view

Soft computing **complements** hard computing.
A good engineer knows *which tool for which problem* — and often uses **both** in one system.

Quick Check — Round 4 (Applications)

[Q1]

Name the 1987 landmark public-transport system that proved fuzzy control at scale, and list two benefits it demonstrated.

[Q2] Match the technique

- | | |
|--|-----------------------------|
| (i) Anti-lock braking rules | (a) Genetic Algorithm |
| (ii) Timetable scheduling | (b) Neural Network |
| (iii) Tumour detection in MRI | (c) Fuzzy Logic |
| (iv) Shortest network route by pheromone | (d) Ant Colony Optimisation |

[Q3]

For 20 cities the TSP has $\approx 6 \times 10^{16}$ tours. What does this number tell you about using exhaustive (hard) search, and what does a GA give you instead?

[Q4]

Give one application where soft computing would be a *wrong* choice.

[A1]

The **Sendai Subway** (Japan, 1987), fuzzy automatic train operation by Hitachi. Benefits: smoother acceleration/braking (passenger comfort), ~10% energy saving, more accurate stopping, reduced wear.

[A2]

(i)–(c) (ii)–(a) (iii)–(b) (iv)–(d)

[A3]

Exhaustive search is **computationally intractable** (NP-hard — centuries of CPU time). A GA abandons the guarantee of optimality and returns a **near-optimal tour in seconds** — exactly Zadeh's trade of precision for *tractability* and *low solution cost*.

[A4]

Any domain requiring provable exactness: bank interest/payroll, cryptographic key computation, avionics arithmetic, compiler code generation. (An *approximate* account balance is unacceptable!)

Summary of Unit 1

What we covered

- ① **Computing systems** — mapping input to output via a precise, unambiguous control action.
- ② **Limits of hard computing** — no model, vague data, intractable problems.
- ③ **Soft computing** — tolerance for imprecision, uncertainty, partial truth and approximation; role model = the human mind.
- ④ **Constituents** — Fuzzy Logic (imprecision), Neural Networks (learning), GA/EC & Swarm (optimisation), Probabilistic Reasoning (uncertainty) — plus *hybrids*.
- ⑤ **Soft vs. Hard** — model-free vs. model-based, approximate vs. exact, robust vs. brittle, complementary rather than competing.
- ⑥ **Characteristics** — human expertise, model-free learning, biological inspiration, fault tolerance, approximation, low cost, goal-driven.
- ⑦ **Applications** — appliances, transport, medicine, vision, finance, power, networks, optimisation.

Soft computing gives machines the power to reason, learn and adapt like humans.

Practice Questions (Exam Oriented)

Short answer (2–5 marks)

- 1 Define soft computing. State its guiding principle.
- 2 What is a control action? Why must it be precise in hard computing?
- 3 Differentiate between imprecision and uncertainty with one example each.
- 4 List any five characteristics of soft computing.
- 5 Why is soft computing said to be a *partnership* and not a single technique?

Long answer (8–10 marks)

- 6 Compare hard and soft computing on at least eight parameters with examples.
- 7 Explain the four constituents of soft computing and the role of each.
- 8 “Soft computing exploits the tolerance for imprecision.” Justify with the car-parking and TSP examples.
- 9 Discuss any four real-life applications of soft computing, naming the technique used and why it suits the problem.
- 10 Explain fault tolerance and graceful degradation. Why does hard computing lack them?

Think & Explore (Take-Home)

Mini design task

Design (on paper) a **fuzzy traffic-signal controller** for a Greater Noida junction:

- Inputs: *queue length*, *waiting time*
- Output: *green-light extension*
- Write 4 linguistic rules
- State why a fixed-timer (hard) controller performs worse

Reading & reflection

- Read Zadeh's 1965 paper abstract, *Fuzzy Sets, Information and Control*.
- Find one recent (2023–2025) research paper applying soft computing in your area of interest — summarise the technique and the gain in one page.
- Prepare for Unit 2: revise *sets*, *relations* and *characteristic functions*.

Reminder

Assignment 1 covers Units 1 & 2 — handwritten, step-by-step, submitted as a single PDF via the course page.

References

Textbooks (as per the course page)

- ① G. J. Klir and B. Yuan, *Fuzzy Sets and Fuzzy Logic: Theory and Applications*, Prentice Hall. **[Main text]**
- ② J.-S. R. Jang, C.-T. Sun and E. Mizutani, *Neuro-Fuzzy and Soft Computing*, Prentice Hall.
- ③ D. E. Goldberg, *Genetic Algorithms in Search, Optimization and Machine Learning*, Addison-Wesley.
- ④ T. J. Ross, *Fuzzy Logic with Engineering Applications*, McGraw Hill.

Seminal papers

- ⑤ L. A. Zadeh, "Fuzzy Sets," *Information and Control*, 8(3):338–353, 1965.
- ⑥ L. A. Zadeh, "Fuzzy Logic, Neural Networks, and Soft Computing," *Communications of the ACM*, 37(3):77–84, 1994.
- ⑦ J. H. Holland, *Adaptation in Natural and Artificial Systems*, 1975.

NPTEL: *Soft Computing*, IIT Kharagpur

Course page: <https://www.gcjana.in/courses/shardauniversity/2601/soft-computing/>

Thank You

Any Questions?

*“So far as the laws of mathematics refer to reality, they are not certain;
and so far as they are certain, they do not refer to reality.”*

— Albert Einstein (*the epigraph Zadeh chose for his 1965 fuzzy sets work*)

Post your doubts on the course page →

<https://www.gcjana.in/courses/shardauniversity/2601/soft-computing/#Post-doubts>